

Group Assignment 2 - Group Lab Activity 2

TNE10006/TNE60006 S1 2025

Assignment Weight:

7.5%

Assignment Points:

75

Submission Due Date:

Week 12 Lab session.

Reference Material:

[Sample Final Skills Exam A](#)

Instructions:

1. Form a group of 3-4 people amongst the students present in the lab session.
2. Discuss and answer the questions in Group Assignment 2 with your group members.
3. Organise for your group to meet as needed to complete all the questions.
4. Each group will submit one completed Group Assignment 2
5. Submit Group Assignment 2, in the Canvas shell, under the Group Lab Activity 2
6. Late penalties will apply for submission after the due date.

Group Assignment 2 Sections:

Section 1: Sample Final Skills Exam A – Topology Analysis (30 marks)

Section 2: Sample Final Skills Exam A – Configuration (25 marks)

Section 3: Sample Final Skills Exam A – Troubleshooting (20 marks)

IMPORTANT:

This document has been formatted as a PDF form providing answer fields. You must not convert this file to another file type nor use a PDF editor to insert your answers.

Group Assignment 2 Members Information:

Student Name	Student ID

Section 1: Sample Final Skills Exam A – Topology Analysis (30 marks)

When tasked to build a network end to end, you should first analyse the topology diagram, addressing tables, and other relevant specifications to understand basic network configuration requirements. In this section, you must refer to the information on **Sample Final Skills Exam A** and discuss configuration requirements with your group. The questions below will help guide the topology analysis discussion.

Note: the topology analysis group discussion should be carried out before configuring the network.

Q1. Refer to **Sample Final Skills Exam A** and answer the following questions regarding VLANs, VLAN membership and 802.1q trunking requirements.

- a) How many VLANs must be configured on **Curly**? If any, list the required VLANs specifying VLAN ID and name.
(2 marks)

- b) How many VLANs must be configured on **Moe**? If any, list the required VLANs specifying VLAN ID and name.
(2 marks)

- c) How many access ports must be configured on **Curly**? If any, list the required access ports specifying port ID and VLAN membership.
(1 mark)

- d) How many access ports must be configured on **Moe**? If any, list the required access ports specifying port ID and VLAN membership.
(3 marks)

- e) How many 802.1q trunks must be configured on **Curly**? If any, list the required trunks specifying port ID and switchport mode.
(2 marks)

- f) How many 802.1q trunks must be configured on **Moe**? If any, list the required trunks specifying port ID, switchport mode and native VLAN ID.
(1 mark)

Q2. Refer to **Sample Final Skills Exam A** and answer the following questions regarding inter-VLAN routing requirements.

- a) Use the table to list all interfaces and sub-interfaces that must be configured on **Larry**. For each, specify interface ID, IP address, subnet mask, and 802.1q ID when applicable.
(5 marks)

Interface ID	IP address	Subnet Mask	802.1q ID
<i>Example: Loopback 10</i>	<i>115.23.10.100</i>	<i>255.255.255.240</i>	<i>Not applicable</i>

Note: you might not need to use all rows.

Q3. Refer to **Sample Final Skills Exam A** and answer the following questions regarding switch IP management requirements.

- a) How many VLAN interfaces must be configured on **Curly**? If any, list all required VLAN interfaces specifying interface ID, IP address and subnet mask.
(1 mark)

- b) How many VLAN interfaces must be configured on **Moe**? If any, list all required VLAN interfaces specifying interface ID, IP address and subnet mask.
(1 mark)

- c) Must a default-gateway be configured on **Curly**? If so, specify the default-gateway IP.
(1 mark)

- d) Must a default-gateway be configured on **Moe**? If so, specify the default-gateway IP.
(1 mark)

Q5. Refer to **Sample Final Skills Exam A** and answer the following questions regarding redundancy and port-security requirements.

- a) Must the STP bridge priority be modified on **Curly** and/or **Moe**? If so, list the switches it should be modified on, specifying VLAN(s) and priority value in each case.
(4 marks)

- b) Must an EtherChannel be configured between **Curly** and **Moe**? If so, use the table below to specify the EtherChannel settings.
(2 marks)

Switch	Interface range	Port-Channel ID	Channel-Group Mode
<i>Example</i> S3	<i>G1/0/10-11</i>	<i>Port-Channel 1</i>	<i>Active</i>
Curly			
Moe			

- c) Must Port-Security be configured on **Curly** and/or **Moe**? If so, use the table to specify switch name, interface(s) ID and Port-Security settings
(3 marks)

Switch Name	Interface(s) ID	Max. MAC address allowance	MAC address learning mode	Violation action
<i>Example:</i> S3	<i>Gi1/0/12-24</i>	<i>2</i>	<i>Sticky</i>	<i>Restrict</i>

Note: you might not need to use all rows.

Q6. Refer to **Sample Final Skills Exam A** and answer the following questions regarding end-host IP configuration.

- a) What IPv4 settings can be configured on the **Ethernet PC**?
(1 mark)

IP address:

Subnet Mask:

Default Gateway:

Section 2: Sample Final Skills Exam A – Configuration (25 marks)

Once you have a good understanding of the topology and configuration requirements, you can move on to build the network. For this, you must use Cisco CLI configuration commands on the switches and routers, as well as verification commands to validate the settings. In this section, you will be asked to specify the CLI commands you used to complete **Sample Final Skills Exam A**.

Note: your answers must show CLI commands that include parameters specific to Sample Final Skills Exam A (i.e. generic command sets will not be considered correct answers).

Q1. Refer to **Sample Final Skills Exam A** and answer the following questions regarding global and remote management configuration.

- a) Use the table to list the command(s) used to configure global settings and remote management on each device.
(4 marks)

Setting	Larry	Curly	Moe
Device Hostname			
Banner MOTD			
Disable Domain Lookup			
Synchronous Logging			
Disable Unused Ports			
SSH Management			

Note: if a setting is not required on a device, leave the cell blank or type "Not Required".

- b) What command(s) can be used to verify that all unused ports have been disabled, and that all used ports are enabled?
(0.5 mark)

- c) What command(s) can be used on **Curly** or **Moe** to test SSH remote access to **Larry**?
(0.5 mark)

Q2. Refer to **Sample Final Skills Exam A** and answer the following questions regarding VLANs, VLAN membership and 802.1q trunking configuration.

- a) Use the table to list the commands used on **Curly** and **Moe** to configure VLANs and VLAN membership when applicable.
(2 marks)

Switch Name	VLANs Configuration	VLAN Membership Configuration
Example: S3	<code>S3(config)#vlan 100</code> <code>S3(config-vlan)#name example1</code> <code>S3(config)#vlan 200</code> <code>S3(config-vlan)#name example2</code>	Not applicable
Curly		
Moe		

Note: if VLANs or VLAN membership configuration is not required on a device, leave the cell blank or type "Not Required".

- b) What command(s) can be used to verify VLANs and VLAN membership configuration on **Curly** and **Moe**?
(0.5 mark)

- c) Use the table to list the commands used on **Curly** and **Moe** to configure 802.1q trunk interfaces.
(2 marks)

Switch Name	Trunk Interface(s) Configuration
Curly	
Moe	

- d) What command(s) can be used to verify 802.1q trunking configuration on **Curly** and **Moe**?
(0.5 mark)

Q3. Refer to **Sample Final Skills Exam A** and answer the following questions regarding inter-VLAN routing configuration.

- a) Use the table to list the commands used on **Larry** to configure interfaces, sub-interfaces and loopback interfaces.
(5 marks)

Interface ID	Interface Configuration
<i>Example: Loopback 100</i>	<i>R1(config)#interface loopback 100 R1(config-if)description THIS IS AN EXAMPLE R1(config-if) ip address 115.23.10.10 255.255.128.0</i>

Note: you might not need to use all rows.

- b) What command(s) can be used to verify Inter-VLAN routing configuration on **Larry**?
(1 mark)

Q4. Refer to **Sample Final Skills Exam A** and answer the following questions regarding switch management IP settings configuration.

- a) Use the table to list the commands used on **Curly** and **Moe** to configure management IP settings.
(2 marks)

Switch Name	Management IP Configuration
Curly	
Moe	

- b) What command(s) can be used to verify management IP settings on **Curly** and **Moe**?
(0.5 mark)

Q5. Refer to **Sample Final Skills Exam A** and answer the following questions regarding redundancy and port-security configuration.

- a) Use the table to list the command(s) used to configure redundancy and port-security settings on **Curly** and **Moe**.
(5 marks)

Setting	Curly	Moe
STP		
EtherChannel		
Port-Security		

Note: if a setting is not required on a device, leave the cell blank or type "Not Required".

- c) What command(s) can be used to verify STP settings on **Curly** and **Moe**?
(0.5 mark)
- d) What command(s) can be used to verify EtherChannel settings on **Curly** and **Moe**?
(0.5 mark)
- e) What command(s) can be used to verify port-security settings on **Curly** and **Moe**?
(0.5 mark)

Section 3: Sample Final Skills Exam A – Validation and Troubleshooting (20 marks)

After building a network, you should validate that the network is fully functional and that it meets all specifications. For this, you can use Cisco CLI **show** commands to validate the settings on switches and routers, as well as **ping** commands to test connectivity from each device to all other devices (including a test PC). This process is known as **troubleshooting** and will allow you to detect and fix configuration errors. In this section, you will find **show** outputs and **ping** results based on **Sample Final Skills Exam A**. You must discuss this troubleshooting information with your group to detect configuration errors and specify the commands that must be used to fix them.

Q1. Refer to **Sample Final Skills Exam A** and answer the following questions regarding **physical topology** troubleshooting.

- a) Based on the **show ip interface brief** outputs, are all interfaces in the correct status? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.

(1 mark)

Note: assume all required cables between the devices are in place.

Larry#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/0/1	unassigned	YES	unset	up	up

--More--

Curly#show ip interface brief

Any interface listed with OK? value "NO" does not have a valid configuration

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	unset	administratively down	down
Vlan451	83.174.16.165	YES	manual	up	up
GigabitEthernet0/0	unassigned	NO	unset	administratively down	down
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/3	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/4	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/5	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/6	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/7	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/8	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/9	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/10	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/11	unassigned	YES	unset	up	up
GigabitEthernet1/0/12	unassigned	YES	unset	administratively down	down

--More--

Moe#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	unset	administratively down	down
Vlan451	83.174.16.164	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	down	down
FastEthernet0/3	unassigned	YES	unset	administratively down	down
FastEthernet0/4	unassigned	YES	unset	administratively down	down
FastEthernet0/5	unassigned	YES	unset	administratively down	down
FastEthernet0/6	unassigned	YES	unset	administratively down	down
FastEthernet0/7	unassigned	YES	unset	administratively down	down
FastEthernet0/8	unassigned	YES	unset	administratively down	down
FastEthernet0/9	unassigned	YES	unset	administratively down	down
FastEthernet0/10	unassigned	YES	unset	administratively down	down
FastEthernet0/11	unassigned	YES	unset	administratively down	down
FastEthernet0/12	unassigned	YES	unset	administratively down	down
FastEthernet0/13	unassigned	YES	unset	up	up
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Q2. Refer to **Sample Final Skills Exam A** and answer the following questions regarding **VLANs and VLAN membership** troubleshooting.

- a) Based on the **show vlan brief** output, is the VLANs and VLAN membership configuration correct on **Curly**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.
(2 marks)

Curly#show vlan brief		
VLAN	Name	Status Ports
1	default	active Gi1/0/3, Gi1/0/4, Gi1/0/5 Gi1/0/6, Gi1/0/7, Gi1/0/8 Gi1/0/9, Gi1/0/10, Gi1/0/12 Gi1/0/13, Gi1/0/14, Gi1/0/15 Gi1/0/16, Gi1/0/17, Gi1/0/18 Gi1/0/19, Gi1/0/20, Gi1/0/22 Gi1/0/23, Gi1/1/1, Gi1/1/2 Gi1/1/3, Gi1/1/4
5	Three	active
45	Stooges	active
451	Management	active Gi1/0/21, Gi1/0/24
1002	fddi-default	act/unsup
1003	token-ring-default	act/unsup
1004	fddinet-default	act/unsup
1005	trnet-default	act/unsup

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

- b) Based on the **show vlan brief** output, is the **VLANs and VLAN membership** configuration correct on **Moe**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.

(1 mark)

Moe#show vlan brief		
VLAN	Name	Status Ports
1	default	active Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gi0/1, Gi0/2
5	Three	active Fa0/13, Fa0/14, Fa0/15, Fa0/16
45	Stooges	active Fa0/17, Fa0/18, Fa0/19, Fa0/20
278	Crazy	active
451	Management	active
1002	fddi-default	act/unsup
1003	token-ring-default	act/unsup
1004	fddinet-default	act/unsup
1005	trnet-default	act/unsup

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

Q3. Refer to **Sample Final Skills Exam A** and answer the following questions regarding **802.1q trunking** troubleshooting.

- a) Based on the **show interfaces trunk** output, is 802.1q trunking configured following best practices on **Moe**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.

(1 mark)

Note: assume that EtherChannel settings have not been configured yet.

```
Moe#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	auto	802.1q	trunking	1
Fa0/2	auto	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-4094
Fa0/2	1-4094

Port	Vlans allowed and active in management domain
Fa0/1	1,5,45,278,451
Fa0/2	1,5,45,278,451

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	1,5,45,278,451
Fa0/2	45,278

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

- b) Based on the **show interfaces trunk** output, is 802.1q trunking correctly configured on **Curly**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.

(2 marks)

Note: assume that all required interfaces, on all devices, are enabled and in the up/up status and that EtherChannel settings have not been configured yet.

```
Curly#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi1/0/1	on	802.1q	trunking	1
Gi1/0/2	on	802.1q	trunking	451

Port	Vlans allowed on trunk
Gi1/0/1	1-4094
Gi1/0/2	1-4094

Port	Vlans allowed and active in management domain
Gi1/0/1	1,5,45,278,451
Gi1/0/2	1,5,45,278,451

Port	Vlans in spanning tree forwarding state and not pruned
Gi1/0/1	1,5,45,278,451
Gi1/0/2	5

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

Q4. Refer to **Sample Final Skills Exam A** and answer the following questions regarding **inter-VLAN routing** troubleshooting.

- a) Based on the **show ip route**, **show ip interface brief** and **sh running-config** outputs, are all **Larry** interfaces, sub-interfaces and loopback interfaces configured correctly? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.

(4 marks)

```
Larry#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    83.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C       83.174.0.0/20 is directly connected, GigabitEthernet0/0/1.45
L       83.174.15.0/32 is directly connected, GigabitEthernet0/0/1.45
C       83.174.16.0/25 is directly connected, GigabitEthernet0/0/1.5
L       83.174.16.126/32 is directly connected, GigabitEthernet0/0/1.5
C       83.174.16.128/27 is directly connected, GigabitEthernet0/0/1.278
L       83.174.16.158/32 is directly connected, GigabitEthernet0/0/1.278
    93.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       93.216.87.0/28 is directly connected, Loopback0
L       93.216.87.1/32 is directly connected, Loopback0
```

```
Larry#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0     unassigned      YES unset    administratively down down
GigabitEthernet0/0/1     unassigned      YES unset    up          up
Gi0/0/1.5                83.174.16.126   YES manual  up          up
Gi0/0/1.45               83.174.15.0     YES manual  up          up
Gi0/0/1.278              83.174.16.158   YES manual  up          up
Serial0/1/0              unassigned      YES unset    administratively down down
Serial0/1/1              unassigned      YES unset    administratively down down
GigabitEthernet0         unassigned      YES unset    administratively down down
Loopback0                93.216.87.1     YES manual  up          up
```

```
Larry#show running-config | section interface
interface Loopback0
  ip address 93.216.87.1 255.255.255.240
interface GigabitEthernet0/0/0
  no ip address
  shutdown
  negotiation auto
interface GigabitEthernet0/0/1
  no ip address
  negotiation auto
interface GigabitEthernet0/0/1.5
  encapsulation dot1Q 50
  ip address 83.174.16.126 255.255.255.128
interface GigabitEthernet0/0/1.45
  encapsulation dot1Q 45
  ip address 83.174.15.0 255.255.240.0
interface GigabitEthernet0/0/1.278
  encapsulation dot1Q 278
  ip address 83.174.16.158 255.255.255.224
--More--
```

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

Q5. Refer to **Sample Final Skills Exam A** and answer the following questions regarding **switch management IP settings** troubleshooting.

- a) Based on the **show ip interface brief** outputs, is the VLAN interfaces configuration correct on both **Curly** and **Moe**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.

(1 mark)

```
Curly#show ip interface brief
Any interface listed with OK? value "NO" does not have a valid configuration

Interface          IP-Address      OK? Method Status              Protocol
Vlan1              unassigned     YES unset  administratively down down
Vlan451            83.174.16.165  YES manual  up                  up
GigabitEthernet0/0 unassigned     NO  unset  administratively down down
GigabitEthernet1/0/1 unassigned     YES unset  up                  up
GigabitEthernet1/0/2 unassigned     YES unset  up                  up
--More--
```

```
Moe#show ip interface brief
Interface          IP-Address      OK? Method Status              Protocol
Vlan1              unassigned     YES unset  administratively down down
Vlan278            83.174.16.156  YES manual  up                  up
Vlan451            83.174.16.164  YES manual  up                  up
FastEthernet0/1    unassigned     YES unset  up                  up
FastEthernet0/2    unassigned     YES unset  up                  up
FastEthernet0/3    unassigned     YES unset  administratively down down
--More--
```

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

- b) Based on the **show ip interface brief** and **show ip default-gateway** outputs, are the management IP settings correct on both **Curly** and **Moe**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it. (1 mark)

```
Curly#show ip interface brief
Any interface listed with OK? value "NO" does not have a valid configuration

Interface                IP-Address      OK? Method Status              Protocol
Vlan1                    unassigned      YES unset  administratively down down
Vlan451                  83.174.16.165   YES manual  up                  up
GigabitEthernet0/0       unassigned      NO  unset  administratively down down
GigabitEthernet1/0/1     unassigned      YES unset  up                  up
GigabitEthernet1/0/2     unassigned      YES unset  up                  up
--More--
```

```
Curly#show ip default-gateway
0.0.0.0
```

```
Moe#show ip interface brief

Interface                IP-Address      OK? Method Status              Protocol
Vlan1                    unassigned      YES unset  administratively down down
Vlan451                  83.174.16.164   YES manual  up                  up
FastEthernet0/1          unassigned      YES unset  up                  up
FastEthernet0/2          unassigned      YES unset  up                  up
FastEthernet0/3          unassigned      YES unset  administratively down down
--More--
```

```
Moe#show ip default-gateway
83.174.16.166
```

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

Q6. Refer to **Sample Final Skills Exam A** and answer the following questions regarding **redundancy and port-security** troubleshooting.

- a) Based on the **show spanning-tree** outputs, is the STP configuration for the **Stooges VLAN** correct on both **Curly** and **Moe**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.
(2 marks)

```
Curly#show spanning-tree vlan 45
```

```
VLAN0045
```

```
Spanning tree enabled protocol rstp
```

```
Root ID    Priority    32813
           Address    501c.b070.7f80
           Cost        19
           Port        1 (GigabitEthernet1/0/1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32813 (priority 32768 sys-id-ext 45)
           Address    b4de.31e7.2600
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi1/0/1	Root	FWD	19	128.1	P2p Peer(STP)
Gi1/0/2	Altn	BLK	19	128.2	P2p Peer(STP)
Gi1/0/11	Desg	FWD	4	128.11	P2p

```
Moe#show spanning-tree vlan 45
```

```
VLAN0045
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    32813
           Address    501c.b070.7f80
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32813 (priority 32768 sys-id-ext 45)
           Address    501c.b070.7f80
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

- b) Based on the **show running-config** and **show etherchannel summary** outputs, is the EtherChannel configuration correct on both **Curly** and **Moe**? If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it. (2 marks)

```
Curly#show running-config interface gigabitEthernet 1/0/1
Building configuration...

Current configuration : 90 bytes
!
interface GigabitEthernet1/0/1
  switchport mode trunk
  channel-group 1 mode active
end
```

```
Curly#show running-config interface gigabitEthernet 1/0/2
Building configuration...

Current configuration : 124 bytes
!
interface GigabitEthernet1/0/2
  switchport trunk native vlan 451
  switchport mode trunk
  channel-group 1 mode active
end
```



```
Moe#show running-config interface fastEthernet 0/1
Building configuration...

Current configuration : 85 bytes
!
interface FastEthernet0/1
  switchport mode trunk
  channel-group 1 mode active
end
```

```
Moe#show running-config interface fastEthernet 0/2
Building configuration...

Current configuration : 119 bytes
!
interface FastEthernet0/2
  switchport trunk native vlan 451
  switchport mode trunk
  channel-group 1 mode active
end
```

```
Curly#show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP       Gi1/0/1(P) Gi1/0/2(s)
```

```

Moe#show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Fa0/1(P)   Fa0/2(s)

```

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.

- c) Based on the **show port-security** output, is the port-security configuration correct on **Moe**?
If not, use the table to list the error(s). For each error, specify the configuration command(s) that must be used to fix it.
(3 marks)

```

Moe#show port-security
Secure Port    MaxSecureAddr  CurrentAddr    SecurityViolation  Security Action
              (Count)        (Count)        (Count)
-----
    Fa0/17                1              0              0      Shutdown
    Fa0/18                1              0              0      Shutdown
    Fa0/19                1              0              0      Shutdown
    Fa0/20                1              0              0      Shutdown
    Fa0/21                4              0              0      Restrict
    Fa0/22                4              0              0      Restrict
    Fa0/23                4              0              0      Restrict
    Fa0/24                4              1              0      Restrict
-----
Total Addresses in System (excluding one mac per port)      : 0
Max Addresses limit in System (excluding one mac per port) : 8192

```

Error Description	Re-configuration Command(s)

Note: you might not need to use all rows.

Note: the re-configuration commands must display device name and configuration mode.